NOTE: The document identifier and heading has been changed on this page to reflect that this is a performance specification. There are no other changes to this document. The document identifier on subsequent pages has not been changed, but will be changed the next time the document is revised.

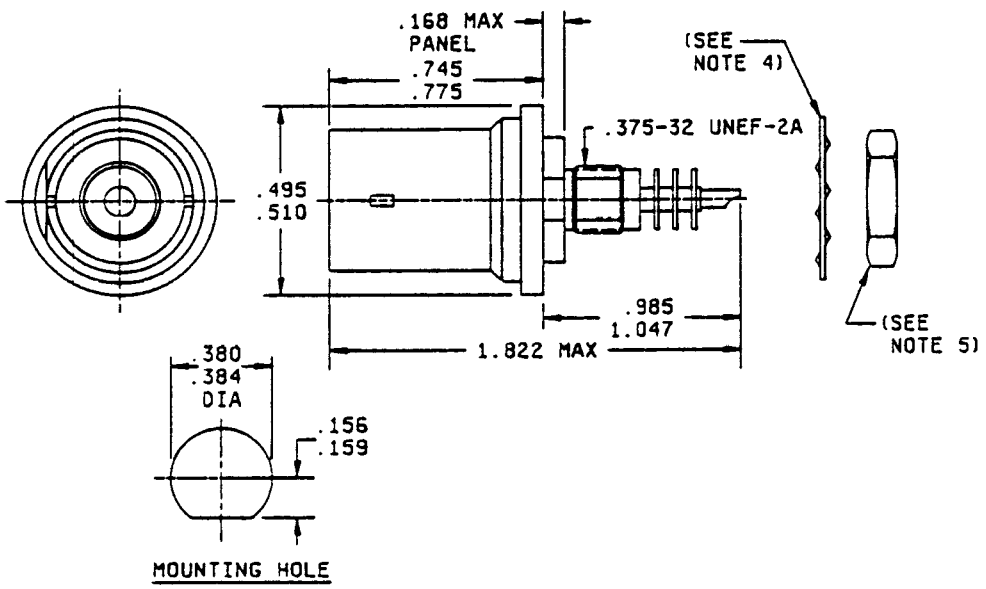


PERFORMANCE SPECIFICATION SHEET

CONNECTORS, COAXIAL, RADIO FREQUENCY, HIGH VOLTAGE
(SERIES SHV (UNCABLED) - RECEPTACLES, PIN CONTACT, JAM NUT MOUNTED, CLASS 2)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the connectors described herein shall consist of this document and the latest issue of Specification MIL-PRF-39012.




NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only and are based upon 1.00 inch = 25.4 mm.
3. All undimensioned pictorial representations are for reference purposes only.
4. Lockwasher, internal, .510 ±.005 inch (12.95 ±.13 mm), maximum O.D. X .022 ±.005 inch (.56 ±.13 mm) thick.
5. Jam nut .500 ±.005 inch (12.70 ±.13 mm) across flats X .125 ±.005 inch (3.18 ± .13 mm) thick.
6. Orientation of the body hex flats, mounting flats and bayonet studs shall be within ±3° of the orientation shown.

FIGURE 1. General configuration.





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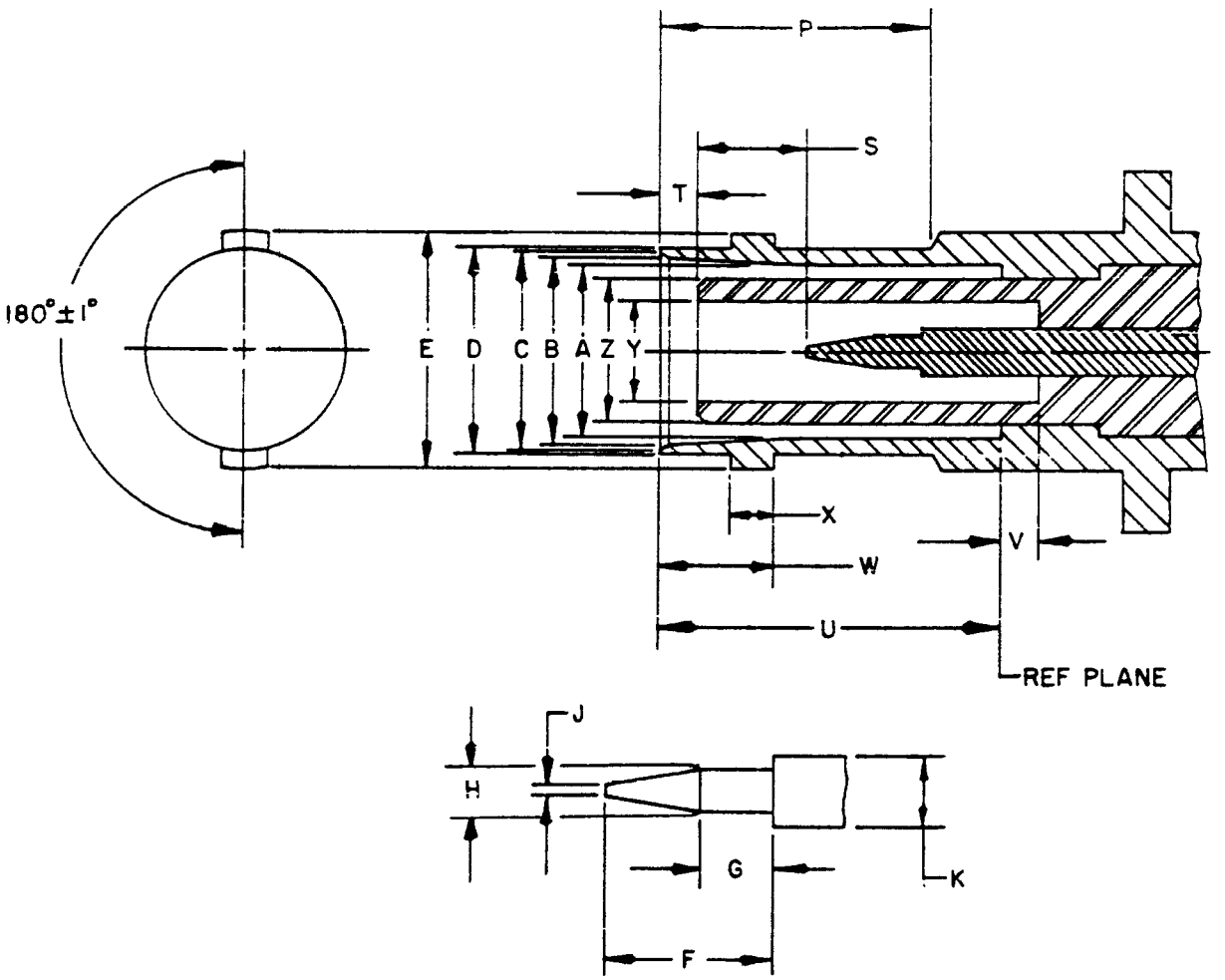



FIGURE 2. Mating dimensions.



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| Ref | Inches | | Millimeters | |
|---|---|---|---|---|
| | Min | Max | Min | Max |
| A | .319 | .321 | 8.10 | 8.15 |
| B | .328 | .333 | 8.33 | 8.46 |
| C | .347 | .357 | 8.81 | 9.07 |
| D | .378 | .382 | 9.60 | 9.70 |
| E | .432 | .436 | 10.97 | 11.07 |
| F | .207 | .214 | 5.26 | 5.44 |
| G | .130 | | 3.30 | |
| H | .052 | .054 | 1.32 | 1.37 |
| J | .015 | .025 | .38 | .64 |
| K | .081 | .083 | 2.06 | 2.11 |
| P | .427 | | 10.85 | |
| S | .188 | .208 | 4.78 | 5.28 |
| T | .061 | .078 | 1.55 | 1.98 |
| U | .626 | .630 | 15.90 | 16.00 |
| V | .064 | .086 | 1.63 | 2.18 |
| W | .204 | .208 | 5.18 | 5.28 |
| X | .075 | .081 | 1.90 | 2.06 |
| Y | .190 | .196 | 4.83 | 4.98 |
| Z | | .260 | | 6.60 |

NOTES:
1. Metric equivalents are given for general information only and are based upon 1.00 inch = 25.4 mm.
2. .005 (.13 mm) flat permissible to meet dimension W.

FIGURE 2. Mating dimensions - Continued.



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ENGINEERING DATA:

Nominal impedance: Nonconstant.

Frequency range: Not applicable.

Voltage rating:

3,500 volts rms and 5,000 Vdc, maximum working voltage at sea level.
350 volts rms, maximum at 70,000 feet.

Temperature rating: -65° to +165°C.

REQUIRMENTS:

Design and construction: See figure 1.

Force to engage and disengage:

Longitudinal force - 3 pounds, maximum.
Torque - 2 1/2 inch-pounds, maximum.

Coupling proof torque: Not applicable.

Inspection conditions: Coupling torque - Not applicable.

Mating characteristics: See figure 2 for dimensions.

Center contact retention: 6 pounds minimum axial force.

Hermetic seal: Not applicable.

Leakage (pressurized connectors): Not applicable.

Insulation resistance: Method 302, test condition B, MIL-STD-202. 1 x $10^{12}$ ohms, minimum.

Corrosion (salt spray): Method 101, test condition B, MIL-STD-202.

Connector durability:

500 cycles minimum at 12 cycles/minute, maximum.
The connector shall meet the mating characteristics and force to engage and disengage requirements.

Contact resistance: In milliohms maximum.

| | Initial | After environment |
|---|---|---|
| Center contact | 2.1 | 2.5 |
| Outer contact | 1.5 | Not applicable |
| Braid to body | .1 | Not applicable |

Dielectric withstanding voltage: Method 301 of MIL-STD-202. 5,000 volts rms and 10,000 Vdc, minimum at sea level.

Vibration, high frequency: Method 204, test condition A, MIL-STD-202. No discontinuity permitted.

Shock: Method 213 of MIL-STD-202, test condition I. No discontinuity permitted.

Thermal shock : Method 107, test condition B, MIL-STD-202, except test high temperature shall be +85°C. High temperature shall be +200°C for connectors using +200°C cable.



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Humidity: Method 103, test condition B per MIL-STD-202. No measurements at high humidity. Insulation resistance shall be at least $10^{12}$ ohms within 5 minutes after removal from humidity.

Corona level:

Voltage - 500 volts, minimum.
Altitude - 70,000 feet.

RF high potential withstanding voltage:

Voltage and frequency - Not applicable.
Leakage current - Not applicable.

Cable retention force: Not applicable.

Coupling mechanism retention force: Not applicable.

Part number: M39012/110- (dash number from table I).

TABLE I. Dash number and applicable cable.

| Dash no. 4/ | Applicable cable no. 3/ |
|---|---|
| CATEGORY A - Field serviceable (no special tool required) | |
| 1/ 0001 (Contacts with gold finish) | M17/29 - RG59 *<br>M17/30 - RG62<br>M17/90 - RG71<br>M17/97 - RG210 |
| 2/ 0002 (Contacts with silver finish) | M17/29 - RG59 *<br>M17/30 - RG62<br>M17/90 - RG71<br>M17/97 - RG210 |

NOTES:
* Cable to be used when performing tests requiring cable.
1/ Gold plated contacts: Minimum thickness of 0.00005 inch in accordance with MIL-G-45205, type II, class I with suitable underplate.
2/ Silver plated contacts: Minimum thickness of 0.0002 inch.
3/ See MIL-C-17 for working voltage of cables.
4/ Panel thickness: .045 inch (1.14 mm) minimum, .168 inch (4.27 mm) maximum.

Custodians:
Army - CR
Navy - EC
Air Force - 85

Review activities:
Army - MI
Navy -
Air Force - 11, 17, 99
DLA - ES

User activities:
Army - AR, AT, ME
Navy - AS, MC, OS
Air Force - 19

Preparing activity:
Army - CR

Agent:
DLA - ES

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